Community Aware Anomaly Detection in Weighted Networks

Social Network Analysis for Computer Scientists — Course paper

Nael Belhaj Haddou
n.t.j.r.belhaj.haddou@umail.leidenuniv.nl
LIACS, Leiden University

Yutao Liu
y.liu.67@umail.leidenuniv.nl
LIACS, Leiden University

ABSTRACT

Detecting anomalies in large weighted networks is a critical challenge in identifying fraud and irregular behaviors. While classical local methods effectively utilize edge weights, they often miss global structural deviations. Conversely, community-aware methods capture global inconsistencies but typically neglect interaction intensities. This paper bridges this gap by proposing **WCADA**, a weighted extension of the Community-Aware Detection of Anomalies (CADA) algorithm. We validate our approach on synthetic weighted LFR benchmarks and the real-world Enron email dataset. Our results demonstrate that WCADA significantly improves anomaly ranking compared to the unweighted baseline (e.g., improving the rank of known anomalies from ≈3400 to ≈1100 in the Enron dataset). Furthermore, we reveal that local and community-aware methods identify largely distinct sets of anomalies (Jaccard similarity ≈ 0). Leveraging this orthogonality, we introduce a hybrid ensemble strategy that combines local weighted features and global community structure to maximize detection coverage.

KEYWORDS

anomaly detection, weighted graphs, social network analysis, network science

ACM Reference Format:

Nael Belhaj Haddou and Yutao Liu. 2025. Community Aware Anomaly Detection in Weighted Networks: Social Network Analysis for Computer Scientists — Course paper. In *Proceedings of Social Network Analysis for Computer Scientists Course 2025 (SNACS '25).* ACM, New York, NY, USA, 6 pages.

1 INTRODUCTION

Complex networks are widely used to model interactions in social, financial, biological, and information systems. Detecting anomalous nodes—those that deviate from expected structural patterns—is essential for identifying fraud, intrusions, or irregular behavior across such systems. Accurate detection improves both network security and understanding of underlying dynamics.

The key challenge addressed in this paper is identifying node anomalies in large weighted networks while ensuring that interaction strength is considered alongside structural patterns. Classical methods such as OddBall [1], while effectively integrating weights, are limited by focusing only on local ego-network characteristics and detecting deviations in the power-law relationship between node and edge counts. Although these locally centered methods are efficient, they overlook anomalies that disrupt the global organization of the network, particularly those connecting multiple communities. Therefore, a detection approach that is aware not only of communities but also of connection weights is required.

Recent work has introduced global and community-aware perspectives. In particular, Helling et al. [6] proposed the Community-Aware Detection of Anomalies (CADA) algorithm, which measures how strongly a node connects to multiple communities without clearly belonging to any. By incorporating community structure, CADA captures irregular nodes that local methods often miss and remains parameter-free and scalable.

However, interaction in real-world networks is rarely binary; edges often carry weights representing transaction amounts, communication frequency, or tie strength. Ignoring these weights discards critical information about the anomaly's potential impact.

Addressing this, this paper extends the community-aware framework to weighted graphs. We implement and evaluate this weighted variant against OddBall on both synthetic (weighted LFR benchmark) and real-world datasets, specifically analyzing how incorporating edge weights into community information improves anomaly detection accuracy and scalability compared to unweighted or local-only approaches.

Furthermore, distinct types of anomalies manifest at different scales. While global methods excel at identifying nodes bridging multiple communities, they may overlook strictly local irregularities, such as "heavy" stars or near-cliques hidden within a single community. Conversely, local methods like OddBall capture these ego-network spikes but remain blind to global structural violations. This suggests that a "one-size-fits-all" approach is insufficient. A hybrid strategy that leverages the complementary strengths of local weighted features and global community structure offers a promising direction for maximizing detection coverage and robustness.

To this end, the contributions of this paper are threefold: (1) We propose *WCADA*, a novel extension of the CADA algorithm that explicitly incorporates edge weights into the community connectivity measure, allowing for the detection of anomalies driven by interaction intensity rather than just topology; (2) We empirically demonstrate that this weighted community-aware approach significantly improves detection rankings compared to the original unweighted CADA and identifies global anomalies that local baselines (OddBall) may overlook, validating the necessity of weight integration; (3) We develop and evaluate a hybrid ensemble strategy combining local (OddBall) and global (*WCADA*) scores, analyzing the trade-offs in efficiency and interpretability to provide a comprehensive detection framework.

SNACS '25, Master CS, Fall 2025, Leiden, the Netherlands

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces preliminaries and definitions. Section 4 describes the proposed approach and algorithms. Section 5 presents the datasets used for evaluation. Section 6 reports and discusses the experimental results, and Section 7 concludes the paper.

2 RELATED WORK

Anomaly detection in complex networks has been approached from several methodological directions, ranging from local structural heuristics to global statistical modeling.

Early work predominantly focused on local neighborhood structures, where the assumption is that anomalous behavior manifests within the ego-network of a node. A prominent representative of this line of work is the OddBall algorithm [1], which identifies deviations from power-law relationships between ego-network size, density, weighted connectivity, and eigenvalue patterns. Such methods can effectively reveal extreme star-shaped or near-clique structures, commonly associated with spammers, intruders, or abnormal communication accounts. Later studies expanded this local perspective by introducing alternative ego-network descriptors [7], considering 2-hop or extended neighborhoods [11], or analyzing nodes with unusually low neighbor overlap [8]. These approaches tend to be scalable and interpretable, but, by design, they are limited to anomalies whose irregularities are detectable from local connectivity alone.

A second stream of work incorporates global structural information, recognizing that anomalies may arise from deviations in higher-order network organization. Proximity-based techniques rely on random walks or Personalized PageRank to evaluate whether a node's position is unexpectedly far from typical diffusion flow [13], while low-rank matrix decomposition methods treat the adjacency matrix as a combination of regular global structure and sparse irregularities, identifying anomalies through residual patterns [4]. Embedding-based approaches further map nodes into low-dimensional Euclidean spaces, capturing global structural similarity and flagging nodes with unusual embedding behavior [11]. Although these global approaches extend beyond the limitations of ego-network methods, many require substantial parameter tuning and may lose interpretability or often struggle to effectively differentiate structural anomalies from regular community members, particularly in networks where mesoscopic structures such as communities dominate the overall topology.

Given that many real-world complex networks exhibit strong community structure, a more recent line of research has focused explicitly on community-aware anomaly detection. This perspective is motivated by the observation that some anomalies cannot be detected locally but become evident when considering their connectivity across multiple communities. The Community-Aware Detection of Anomalies (CADA) algorithm [6] operationalizes this idea by quantifying how broadly a node distributes its connections across communities detected via algorithms such as Louvain [3] or Infomap [12]. Nodes that connect to many different communities while not strongly belonging to any single one are flagged as "community-agnostic" nodes.CADA has demonstrated strong performance on LFR benchmark networks [10] and real-world graphs, but it is sensitive to the choice of community detection algorithm and, in its original formulation, does not naturally incorporate weighted edges.

In summary, prior research provides complementary strengths: local methods such as OddBall capture ego-structural irregularities, global methods capture deviations from expected connectivity patterns, and community-aware methods leverage modular structure to detect anomalies invisible to local or global techniques alone. However, relying on a single perspective leaves detection blind spots, and current community-based methods fail to utilize edge weights. To address these gaps, our work extends CADA to weighted graphs and proposes a hybrid strategy that combines local and community-aware scores. We evaluate these approaches against OddBall using standardized weighted LFR benchmarks and real-world data, specifically quantifying the performance gains achieved by integrating weight information and combining detection levels.

3 PRELIMINARIES

This section includes informal definitions necessary to understand the methodology and results. It then builds up to the problem statement of this paper.

3.1 Network Terminology

Let $G = (V, E)$ denote an undirected, weighted network, where V is the set of vertices with size $n = |V|$, and E is the set of edges with size $m = |E|$.

In this work, we explicitly incorporate edge weights to capture interaction intensity. Specifically, each edge $(u, v) \in E$ is associated with a weight:

$$w_{uv} > 0 \tag{1}$$

The degree of a node v, denoted by k_v (or $d(v)$), is defined as the number of edges incident to v, and the average degree of the network is denoted by $\bar{k}$.

An *ego network* for a specific node u (the "ego") consists of the node itself, its direct neighbors $N(u)$, and the connecting edges. Structurally, vertices often cluster into *communities* $C \subset V$, which are subsets more densely connected internally than to the rest of the network [5]. Other key topological properties include sparsity and the power-law degree distribution [2].

3.2 Problem Statement

Formally, the problem addressed in this paper is formulated as follows:

Input: We are given a static, undirected, weighted network $G = (V, E, W)$, where W represents the set of edge weights.

Task: The goal is to derive an anomaly scoring function $f : V \to \mathbb{R}$ that assigns a score y_v to each node $v \in V$.

Objective: The function f should assign higher scores to nodes that exhibit anomalous behavior. In this work, we define a node as anomalous if it violates expected patterns in terms of either:

1. **Global Community Structure:** Nodes that bridge multiple communities without clear membership to any single community (as targeted by community-aware methods).
2. **Interaction Intensity:** Nodes whose weighted connectivity patterns deviate significantly from the expectations set by their local neighborhood or global structural role.

Finally, the problem implies ranking all nodes $v \in V$ based on $f(v)$ such that the top-k ranked nodes correspond to the true anomalies in the network.

4 APPROACH

First, two of the previous approaches used for this type of problem are explained in Section 4.1. Afterwards, the method proposed here is explained in Section 4.2.

4.1 Existing Approaches

4.1.1 OddBall. OddBall [1] primarily focuses on features extracted from the ego networks of nodes. For each node $i \in V$, the algorithm extracts the number of nodes N_i and the number of edges E_i within its ego network. It then fits the Egonet Density Power Law, denoted as $E_i = CN_i^\alpha$. The core premise is that the more a node deviates from this power-law relationship, the more anomalous it is considered.

The anomaly score, denoted as $ob(i)$, is computed as follows:

$$ob(i) = \frac{CN_i^\alpha}{E_i} \log(E_i - CN_i^\alpha + 1) \tag{2}$$

where CN_i^α represents the expected number of edges derived from the power law, and E_i is the actual observed number of edges.

4.1.2 CADA. The Community-Aware Detection of Anomalies (CADA) algorithm [6] is a parameter-free method that assesses a node's anomalousness based on its connectivity across multiple communities.

$$cd(i) = \sum_{j=1}^{c} \frac{g_i^j}{g_i^*} \tag{3}$$

As mentioned, CADA requires a separate community detection algorithm, the authors experiment with Infomap [12] and the Louvain algorithm [3]. Infomap is an information-theoretic method using random walks to infer information flow in the network. On the other hand, the Louvain algorithm is a greedy algorithm that optimizes the modularity function, a measure of the current community's quality. From here on out, the CADA using Infomap and the Louvain algorithm will be referred to as $CADA_I$ and $CADA_L$ respectively.

4.2 Proposed Approaches

4.2.1 CADA Extensions: WCADA. To address the limitation of binary connectivity in the original formulation, we propose a weighted extension of the algorithm, denoted as *WCADA*.

While the original *CADA* evaluates the diversity of a node's connections based on the raw count of edges to different communities, *WCADA* explicitly incorporates edge weights to quantify the *intensity* of these interactions. Let w_{uv} denote the weight of the edge between nodes u and v. For a node i and a community C_j, we define the weighted connectivity strength W_i^j as the sum of weights of edges connecting i to all nodes in C_j:

$$W_i^j = \sum_{v \in N(i) \cap C_j} w_{iv} \tag{4}$$

where $N(i)$ is the set of neighbors of node i.

The anomaly score for *WCADA*, denoted as $cd_w(i)$, is then computed by normalizing these weighted strengths against the node's maximum connection strength to any single community:

$$cd_w(i) = \sum_{j=1}^{c} \frac{W_i^j}{W_i^*} \tag{5}$$

where $W_i^* = \max_j W_i^j$. This modification ensures that a node is flagged as anomalous not just for having links to many communities, but for maintaining *strong* interactions with multiple distinct modules, which is a stronger indicator of structural evasion or bridging roles in weighted graphs.

To evaluate the consistency of our approach and mitigate the sensitivity to the underlying community structure mentioned in previous work [6], we implement *WCADA* using partitions generated by two distinct algorithms: the modularity-based Louvain algorithm [3] and the flow-based Infomap algorithm [12].

4.2.2 Ensemble Methods. To combine the complementary strengths of local and global detection, we employ a linear ensemble strategy. Let $\tilde{ob}(v)$ and $\tilde{cd}(v)$ denote the Min-Max normalized anomaly scores obtained from OddBall and *WCADA*, respectively. The ensemble score $en(v)$ is defined as:

$$en(v) = \alpha \cdot \tilde{ob}(v) + (1 - \alpha) \cdot \tilde{cd}(v) \tag{6}$$

where $\alpha \in [0, 1]$ is a trade-off parameter controlling the relative importance of local versus community-aware features. When $\alpha = 1$, the method relies solely on local OddBall features; when $\alpha = 0$, it depends entirely on global community structure.

5 DATA

5.1 Synthetic Data

Multiple synthetic networks are generated using a simplified weighted version of the Lancichinetti-Fortunato-Radicchi (LFR) benchmark [9]. This is because networks from that benchmark follow many properties of real-world networks closely like the distributions followed by degree, community size and weight. This is also based on the data generation method used by Helling et al. [6], ensuring the experiments are comparable.

The simplified weighted LFR benchmark first generates n nodes and a degree distribution following a power law with parameter γ_1 as exponent and the extremes of the distribution are chosen such that average degree is k. Edges are then formed such that a fraction μ connects a node to another node outside its community. Afterwards, community sizes are generated following another power law, this one with parameter γ_2 as exponent, such that the total size equals the number of nodes.

Once an unweighted network is generated, a weight distribution following another power law k_i^β is generated, with k_i as node i's degree and β a parameter of the algorithm. This distribution is then used to generate the total weight of each node's connections. Finally, the weight of each connection is assigned based on the strength of both nodes and μ_w which controls what fraction of a node's total weight links it to otherr communities.

The generated networks contain between 10,000 nodes. The weight mixing parameter is the fraction of a node's total weight assigned to links to other communities, here values between 0.1 and

0.5 are used. The rest of the data generation parameters is shown in appendix A.

Three types of anomalies are then added to the generated network: The first is random anomalies where a node is generated with a degree at least k drawn from the same distribution as the rest of the graph and connected to random nodes. The second is a set of a nodes that rewire to themselves all edges connected to 2 to 21 nodes per anomaly in the graph and remove the nodes from the graph. The last anomaly type multiplies the weight of one random edge by 50, marking both nodes it connects as anomalies.

5.2 Real World Data

To ensure the algorithms show results in a true setting they are also compared using a real world dataset, the Enron dataset. This dataset contains emails from Enron exchanged between 1998 and 2002. From this dataset, a network with employees and other peope linked to the company as nodes, emails exchanged as edges and number of emails as edge weights can be generated.

6 EXPERIMENTS

This section presents the different experiments that were conducted to study the performance of the algorithms earlier. First the experimental setups are described and the parameters used are made available. Afterwards we present the results of our experiments.

6.1 Experimental Setup

In the first experiment, 4 versions of *CADA* are tested against OddBall and each other on a 10,000 node synthetic network with a mixing parameter of 0.3. Both of the two versions of *CADA* are evaluated with both community detection algorithms. The top 1% most anomalous nodes found are compared using F_1-score which is defined in equation 7 where precision is the ratio of correct and incorrect identifications and recall is the fraction of the added anomalies that were found. In this experiment, OddBall is used as a baseline to understand how the versions of *CADA* compare to an algorithm already used with weighted networks. This gives us insight on whether accounting for edge weights improves performance.

$$F_1 = \frac{precision \times recall}{precision + recall} \tag{7}$$

The second experiment compares versions of the ensemble method using different values of α to the OddBall baseline. This experiment is also performed on synthetic data with 10,000 nodes but has a fixed weight mixing parameter set to 0.2. As this experiment is also conducted on synthetic data, results are once again compared using F_1-scores of their top 1% anomalies. This helps us understand how viable the ensemble model is and what kinds of ratios are credible settings (with these network parameters). Note that for both experiments on synthetic data results are averages of 10 repetitions.

The last experiment is a comparison of OddBall, the two *CADA* versions using Infomap and the an ensemble with α set to 0.4. This experiment is conducted on the Enron dataset, meaning it does not have an explicit ground truth. However, the Enron case has concluded and many of the guilty parties are known. As a consequence the results can be compare qualitatively by comparing the ranking of a few key actors depending on the algorithm used, here we analyze the scores attributed to former chairman and CEO Kenneth Lay, former CEO Jeffrey Skilling and former CFO Andrew Fastow. Additionally, we measure the overlap between each method's top one percent anomalies using Jaccard Similarity as defined in equation 8.

$$J(A_1, A_2) = \frac{|A_1 \cap A_2|}{|A_1 \cup A_2|} \tag{8}$$

6.2 Results

6.2.1 CADA Versions Comparison. The performances of each algorithm can be seen in 1. There does not appear to be a sole clear best performer but OddBall and $WCADA_I$ identify two anomaly types each better than the three other algorithms for all values of μ_w. OddBall appears to have a large advantage over all *CADA* versions when identifying weight anomalies as those fit very well with one of its assumptions. On the other hand as OddBall ignores the community structure it fails to identify the random anomalies.

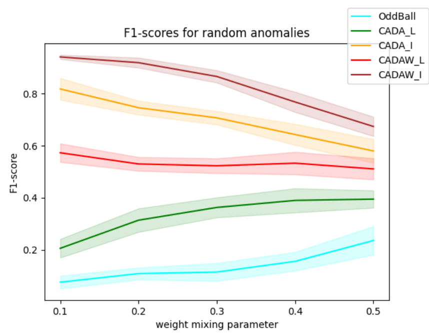
The weight mixing parameter has inconsistent but mostly negative effects on the different versions of *CADA*. This is surprising as the non-weighted *CADA* versions should be completely agnostic to that parameter as they do not consider weight. On the other hand, it does not impair OddBall which even seems to perform slightly better with higher values of μ_w. This is likely caused again by OddBall ignoring the community structure, as μ_w has no impact on the network's topology, it has no influence on OddBall's performance.

6.2.2 Ensemble Method. The performance comparison of different ratios of normalized OddBall and $WCADA_I$ is shown in figure 2. It is immediately apparent that this model inherits qualities from both models that compose it. Any model tested that integrates a part of OddBall's score performs about as well as it does on its own to identify weight anomalies. A similar observation can be made of the ensembles' performance on random anomalies which approaches *CADA*'s best in the previous experiment. Finally, the ensemble outperforms both of the algorithms it is based at identifying replace anomalies. This is likely because those anomalies have unusual neighborhood and edges because of how they are generated while being inconsistent with the larger network's structure.

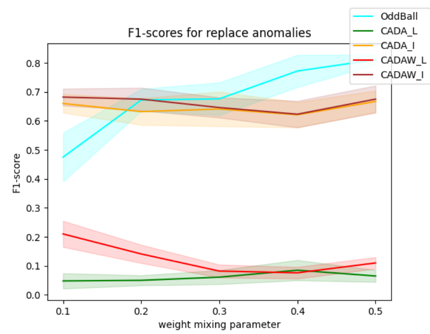
As mentioned above, the ensemble performs poorly on one anomaly type when the coefficients are taken to extremes. However it can be observed that values closer to 0.6 offer the best performances overall by a large margin. The difference in overall performance is less due to an improvement over OddBall and $WCADA_I$ on any one anomaly type and mostly due to it avoiding performance collapse on all anomaly types. This s consistent with the idea behind the ensemble model of benefit from both algorithms' strengths while using one to compensate for the other's shortcomings in one area.

6.2.3 Enron Anomaly Detection. Table 6.2.3 includes the ranking of three important members of the Enron network. The network contains almost 40,000 nodes, this means that all methods find these individuals to be within the top 10% most anomalous. OddBall and the ensemble rank 2 of the 3 individuals much higher (6 to more than 10 times higher) than both version of *CADA* do.

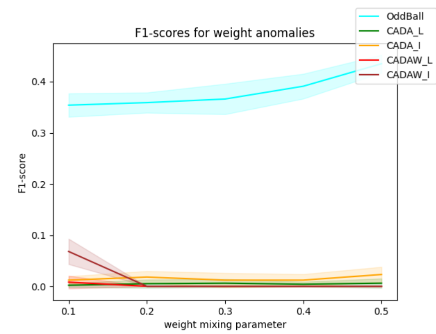
Contradicting the poor performance of the unweighed *CADA* versions. $CADA_I$ is by far the best at identifying Andrew Fastow.



(a) 1a

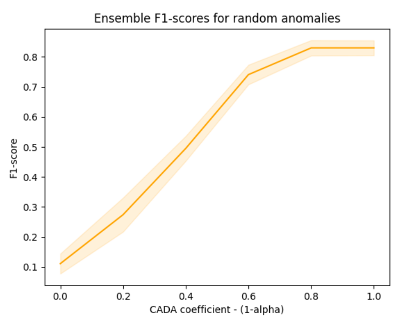


(b) 1b

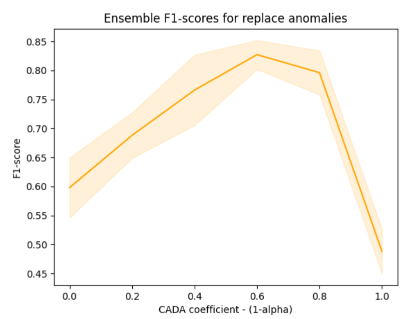


(c) 1c

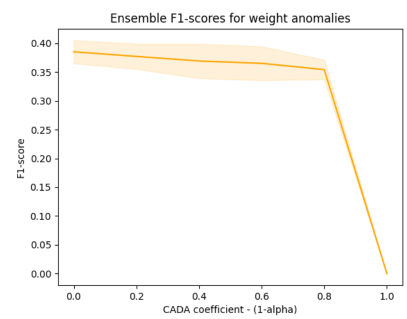
Figure 1: Plots of the performance of 4 versions of *CADA* and OddBall at detecting different types of anomalies in a synthetic network. The performance is measured by F1-scores of the actual node anomalies and the algorithm's top 1% and larger values indicate better performance. OddBall and $WCADA_I$, both perform very well on replace anomalies but OddBall underperforms on random anomalies. All versions of *CADA* perform very poorly on weight anomalies. But the ones ignoring weights also perform poorly with replace anomalies. The weight mixing parameter seems to have a small but inconsistent influence.



(a) 2a



(b) 2b



(c) 2c

Figure 2: Plots of the F1-scores of different ensemble models using $WCADA_I$ and OddBall with different coefficients and on (a) random anomalies (b) replace anomalies and (c) weight anomalies. The models with the highest *CADA* coefficients perform well on random anomalies but quite poorly on both other types. On the other hand, models with high OddBall coefficients perform well with weight anomalies but poorly on both other types. A coefficient of 0.6 appears to produce the best performance when considering all anomaly types.

This could be explained by the role of CFO requiring him to contact many departments rarely but communicating far more with a set of other employees. The number of edges to different communities within the company would make this anomalous to an unweighted *CADA* but the large number of emails within his community obfuscates him from algorithms accounting for weight.

	OddBall	$CADA_I$	$WCADA_I$	Ensemble
Kenneth Ley	247	3449	1145	247
Andrew Fastow	3345	296	1949	3199
Jeffrey Skilling	262	3790	1803	258

Table 1: Ranks of the anomalousness of email communications conducted by three key characters of the Enron case according to different algorithms.

Looking at the Jaccard similarities of the most anomalous nodes found by each method explains why they both rank Kenneth Ley and Jeffrey Skilling so similarly. As shown in table 6.2.3, OddBall and the ensemble share the exact same set of nodes as their top 1% although their rankings differ. This indicated that the distribution of scores in OddBall is much larger here. As the α coefficient is set to 0.4, thus both scores are almost weighed equally. This shows that to be so overwhelming in the ensemble after being normalized the OddBall scores must have a very different distribution.

	OddBall	$CADA_I$	$WCADA_I$	Ensemble
OddBall	–	0	0	1
$CADA_I$	0	–	0.28	0
$WCADA_I$	0	0.28	–	0
Ensemble	1	0	0	–

Table 2: Jaccard similarities of the top 1% most anomalous nodes from the Enron dataset by OddBall, two versions of *CADA* and an ensemble model using OddBall and $WCADA_I$.

7 CONCLUSION

We aimed to adapt *CADA* to weighted networks in order to compare it to OddBall in the setting it was built for. This is accomplished in the first experiment which shows that *CADA* can be used in undirected weighted networks and that scaling the vector it uses to score nodes with weight improves performance significantly. The objective after this was to start work on a model that would combine local and global structure awareness. To that end we used an ensemble method to combine the strengths of both algorithms. The ensemble of $WCADA_I$ and OddBall showcased in the second experiment showed that this method allows us to compensate very well for the weakness of one algorithm while sacrificing little performance.

While the ensemble model is an interesting step towards combining the advantages of both OddBall and *CADA*. The study done here is too small in scale to give results that could be used as is. It would yield interesting insights to observe how the performance of different ratios of ensemble varies with parameters like μ_t or γ_1 and γ_2.

Beyond the empirical improvements, our findings highlight an important conceptual insight: node anomalies in complex networks are inherently multi-scale phenomena. Local ego-network irregularities and global community-bridging behavior capture fundamentally different types of abnormality, and neither perspective alone is sufficient. By extending community-aware detection to weighted networks and empirically demonstrating the orthogonality between local and global methods, this work supports the view that robust anomaly detection should integrate multiple structural lenses.

Future work can build on this by exploring adaptive or data-driven ensemble strategies, specifically to automate the selection of the trade-ff parameter, α, as well as extending the framework to temporal or dynamic weighted networks to capture evolving anomalies.

ACKNOWLEDGMENTS

REFERENCES

[1] Leman Akoglu, Mary McGlohon, and Christos Faloutsos. 2010. oddball: Spotting Anomalies in Weighted Graphs. In *Proceedings of the 14th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*. Springer, Berlin, Heidelberg, 410–421. doi:10.1007/978-3-642-13672-6_45
[2] Albert-László Barabási and Réka Albert. 1999. Emergence of scaling in random networks. *Science* 286, 5439 (1999), 509–512.
[3] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment* 2008, 10 (2008), P10008.
[4] Varun Chandola, Arindam Banerjee, and Vipin Kumar. 2009. Anomaly Detection: A Survey. *Comput. Surveys* 41, 3 (2009), 1–58.
[5] Michelle Girvan and Mark EJ Newman. 2002. Community structure in social and biological networks. *Proceedings of the National Academy of Sciences (PNAS)* 99, 12 (2002), 7821–7826.
[6] Thomas J. Helling, Johannes C. Scholtes, and Frank W. Takes. 2019. A Community-Aware Approach for Identifying Node Anomalies in Complex Networks. In *International Conference on Complex Networks and Their Applications*. Springer, Cham, 244–255. doi:10.1007/978-3-030-05411-3_20
[7] Keith Henderson, Brian Gallagher, Leman Akoglu, Tina Eliassi-Rad, and Hanghang Tong. 2012. It's Who You Know: Graph Mining Using Local Structural Features. *Proc. IEEE* 100, 9 (2012), 1649–1673.
[8] Ming Jiang, Peng Cui, and Christos Faloutsos. 2016. Suspicious Behavior Detection: Current Trends and Future Directions. *IEEE Intelligent Systems* 31, 1 (2016), 31–39.
[9] Andrea Lancichinetti and Santo Fortunato. 2009. Benchmarks for testing community detection algorithms on directed and weighted graphs with overlapping communities. *Phys. Rev. E* 80 (Jul 2009), 016118. Issue 1. doi:10.1103/PhysRevE.80.016118
[10] Andrea Lancichinetti, Santo Fortunato, and Federico Radicchi. 2008. Benchmark graphs for testing community detection algorithms. *Physical Review E* 78, 4 (2008), 046110.
[11] Bryan Perozzi, Rami Al-Rfou, and Steven Skiena. 2014. DeepWalk: Online Learning of Social Representations. In *Proceedings of the 20th ACM SIGKDD Conference*. ACM, New York, NY, USA, 701–710.
[12] Martin Rosvall and Carl T Bergstrom. 2008. Maps of random walks on complex networks reveal community structure. *Proceedings of the National Academy of Sciences (PNAS)* 105, 4 (2008), 1118–1123.
[13] Hanghang Tong, Christos Faloutsos, and Jie Pan. 2006. Fast Random Walk with Restart and Its Applications. In *ICDM*. IEEE, Los Alamitos, CA, USA, 613–622.

A WEIGHTED LFR PARAMETERS

n	$10,000$
k	$2n^{1.15}/n$
$kmax$	$n^{\frac{1}{\gamma_1 - 1}}$
γ_1	3
γ_2	2
μ_t	0.3
μ_w	0.1 to 0.5
β	1.5

Table 3: Parameters used to generate the synthetic datasets used in the experiments